

Multiple Choice

1. **Answer:** D

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: D - False, True

2. **Answer:** C

Options: A) $P(A | B) * P(B | C) * P(C | A)$; B) $P(C | A, B) * P(A) * P(B)$; C) $P(A, B | C) * P(C)$; D) $P(A | B, C) * P(B | A, C) * P(C | A, B)$

Note: Correct option: C - $P(A, B | C) * P(C)$

3. **Answer:** D

Options: A) To save computing time during testing; B) To save space for storing the Decision Tree; C) To make the training set error smaller; D) To avoid overfitting the training set

Note: Correct option: D - To avoid overfitting the training set

4. **Answer:** B

Options: A) they are biased; B) they have high variance; C) they are not consistent estimators; D) None of the above

Note: Correct option: B - they have high variance

5. **Answer:** D

Options: A) Increase bias; B) Decrease bias; C) Increase variance; D) Decrease variance

Note: Correct option: D - Decrease variance

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '2'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Lower variance'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'False, True'.

Long Form Response

11. **Answer:** `def find_lists(Input): | return 1 | return len(Input)`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def find_max_val(n, x, y): | return (ans if (ans >= 0 and`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key